

**Regression-Based Modelling of Glioma Subtype Distributions from
Multi-Modal Magnetic Resonance Imaging**

Final Project Report

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Table of Contents

Table of Contents	1
1. Abstract	2
2. Introduction and Goal	2
3. Dataset Overview	3
4. Exploratory Data Analysis	4
5. Data Pre-Processing	5
1. Train-Test Split	5
2. Normalization	5
3. Labeling Actual Proportions	5
4. Data Loading	6
5. Data Caching	6
6. Modeling Approach & Hyperparameter Tuning	6
1. Whole-Image Regression Pipeline	6
2. Model Architecture: Modified VGG-16	7
3. Training the Base Model & Manual Fine-Tuning	7
4. Base Model by Grid-Search + Fine-Tuning Summary	7
7. Model Evaluation & Results	8
1. Training and Validation Loss Curves	9
2. Test Set Evaluation	9
3. Visual Demonstration	10
8. Code and Resource Organization	11
1. Notebook Versioning through Duplicates	11
2. Modular Code Structure	11
3. Dataset Caching for Efficiency	11
4. Cloud-Based Resource Management	11
9. Challenges & Problem Solving	12
10. Future Direction & Learning Outcomes	12
11. Individual Contributions & Reflection	13
References	14

1. Abstract

Brain tumors arise from uncontrolled cell growth that forms masses within the brain. Depending on their size, location, and stage of development, these tumors can pose significant health risks. Some common symptoms of brain tumors often include: headaches, nausea, vomiting, vision changes, and can affect sensory-motor skills. Gliomas are primary brain tumors that arise from glial cells, which are supportive and protective cells of the brain and spinal cord. High-grade gliomas, especially glioblastomas, are typically aggressive, with an average survival time of 12-18 months, and only 25% of patients survive more than one year. This is important because even non-cancerous tumors can disrupt brain function by compressing vital areas. Therefore, early detection is critical for improving outcomes, making it essential to understand tumor composition as a part of diagnosis and treatment planning. For our approach, we utilized the Brain Tumor Segmentation (BraTS) 2020 dataset, which includes multi-modal magnetic resonance imaging (MRI) scans of high-grade and low-grade gliomas. We built a model to predict the proportions of edema, necrotic core, and enhancing tumor components in individual patients. Our project aims to predict the proportions of each brain tumor component to help better understand tumor composition in individual patients. This provides a quantitative view of tumor composition, supporting more informed, personalized clinical decisions.

2. Introduction and Goal

Brain tumor research is important because it helps doctors find and diagnose tumors earlier and more accurately. It also leads to better treatments and improves patients' lives. By studying how brain tumors work and flow, scientists can create new therapies like targeted treatments, immunotherapies, and gene therapies. Using the tool of our model, this can help doctors verify their findings in brain tumor scans. Research also helps to understand and reduce the side effects of current treatments, making life better for patients. The brain tumor components that we are classifying into proportions include the Necrotic core (1), Edema (2), and Enhancing tumor (4). The Necrotic core is the central region of cell death within a tumor. This is typically not dangerous on its own, but it can contribute to tumor growth and eventually spread to other parts of the brain. Edema is fluid accumulation in the tissue around the tumor. This can be serious and potentially life-threatening when there is an abnormal amount of fluid, which causes swelling and puts pressure on other parts of the brain. Enhancing tumors are regions that show contrast enhancement, indicating an active tumor and tissue that is capable of growing. This is typically the most dangerous component because it highlights the abnormal blood vessel growth in the brain. We also have the MRI modalities that are included in each patient's brain scans that highlight specific regions of the brain. The different MRI modalities include T1, T1CE, T2, and FLAIR, which will be described under the dataset overview. Our project goal is to predict the proportions of each brain tumor component to help better understand tumor composition for individual patients.

3. Dataset Overview

The dataset used in this project originates from the Multimodal Brain Tumor Segmentation Challenge (BraTS) 2020, hosted by the Center for Biomedical Image Computing and Analytics (CBICA) at the University of Pennsylvania. It is a widely recognized benchmark for brain tumor segmentation, classification, and survival prediction tasks.

We accessed the data via Google Cloud Storage, using the public bucket “csc-509-image-files” linked to the “sfsu-378805” project. The bucket was mounted in our Google Colab environment using gcsfuse, allowing us to directly stream and process the image files without manual downloads.

Overview

- Type: Multi-modal 3D MRI scans of glioma patients
- Format: NIfTI (.nii.gz)
- Original Split: Training: 369 patients, Validation: 125 patients

MRI Modalities (per patient)

- T1 – Baseline anatomical scan
- T1CE – T1 with contrast (highlights tumors and blood vessels)
- T2 – Emphasizes edema and fluid content
- FLAIR – Similar to T2, but suppresses cerebrospinal fluid to better show lesions
- Segmentation Mask (seg) – Expert-annotated ground truth labels
 - **Necrotic Core (1)**: Necrotic/non-enhancing tumor core
 - **Edema (2)**: Peritumoral edema
 - **Enhancing Tumor (4)**: GD-enhancing tumor

Metadata

- Patient ID
- Tumor Grade (HGG or LGG)
- Survival Time (in days)
- Age
- Extent of Resection

Preprocessing (Applied by BraTS Organizers)

- Skull-stripping – Removal of non-brain tissue
- Co-registration – Aligned to a common anatomical template
- Resampling – Standardized to $240 \times 240 \times 155$ voxels at 1 mm³ resolution
- Intensity normalization – Applied across all modalities to mitigate scanner variation

4. Exploratory Data Analysis

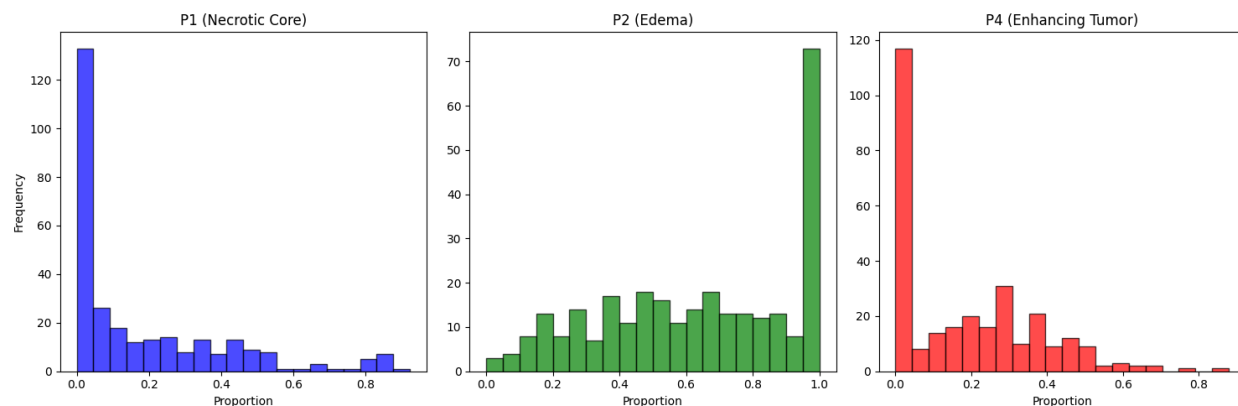


Figure 1. Histograms of actual brain tumor component proportions

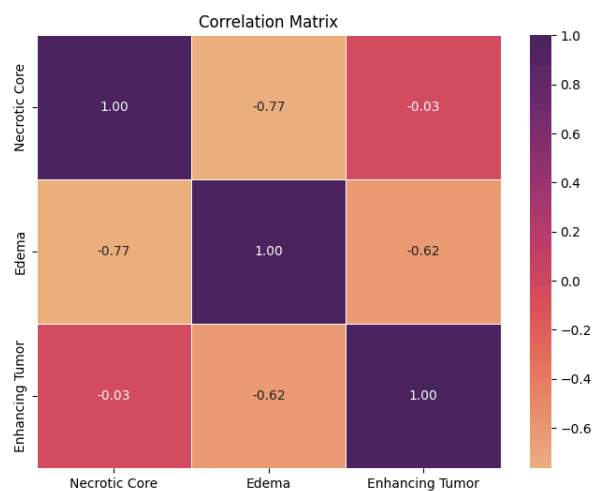


Figure 2. Correlation matrix

	p_1	p_2	p_4
Mean	0.178729	0.638349	0.182922
Median	0.070692	0.655935	0.150464
Std. Dev.	0.228724	0.290397	0.186987
Min	0.0	0.0	0.0
Max	0.922570	1.000000	0.879921

Table 1. Descriptive statistics of brain tumor components

Findings

Based on these figures and the table, it can be observed that Edema (p_2) dominated the tumor composition, with many cases showing near-complete p_2 involvement. This refers to the idea that among the patients who do have Edema, this component accounts for the majority of the tumor's volume. Another observation that can be made is that Necrotic Core (p_1) and Enhancing Tumor (p_4) were right-skewed in Figure 1, with most patients showing little to no presence of these components. One final observation is the moderate to strong negative correlation between Edema and the other tumor components (Figure 2). This indicates that as the presence of Edema increases, the presence of the other two components tends to decrease.

5. Data Pre-Processing

1. Train-Test Split

Because the original BraTS dataset was provided for a competition, we only had access to training and validation sets. To fairly evaluate our model on unseen data, we created our own test set. We combined all available patients, shuffled them, and then split the dataset into 80% training, 10% validation, and 10% test. Below is a breakdown of our split:

- Number of training samples: 294
- Number of training batches: 37
- Number of validation samples: 37

Since we were working with file paths, we saved the split information into a CSV file (split_log.csv) in Google Drive. This allowed us to consistently reference the correct image locations during data loading.

2. Normalization

Normalization standardizes the range of pixel values across all images, which helps improve training stability and model performance. In our case, we used percentile-based min-max normalization to scale pixel values between 0 and 1.

3. Labeling Actual Proportions

While the dataset included segmentation masks as ground truth labels, our model is designed to predict the proportions of each tumor subtype, rather than perform pixel-wise segmentation. To generate ground truth labels suitable for our task, we calculated the proportion of each tumor subtype in the middle axial slice by counting the number of pixels belonging to each label. All segmentations were manually annotated by up to four expert raters and reviewed by board-certified neuroradiologists, making them reliable for this transformation (Multimodal Brain Tumor Segmentation Challenge, n.d.).

4. Data Loading

We implemented a custom PyTorch ‘Dataset’ class to efficiently load and preprocess the data. This class includes the following components:

- **__init__**: Filters the patient directories to include only those that contain a segmentation file, ensuring that the dataset is suitable for supervised learning.
- **__len__**: Returns the total number of valid patient samples in the dataset.
- **__getitem__**: Handles loading of the 3D MRI volumes for all four modalities (T1, T1CE, T2, and FLAIR), extracting the middle axial slice. If the slice contains no tumor (i.e., all segmentation values are 0), the sample is skipped. The four modality slices are stacked, normalized, and resized to 224x224 to match the input size expected by VGG16.

5. Data Caching

Initially, we skipped caching and loaded data on the fly during model training. However, we found that training for just 10 epochs took an average of 2 hours, which was inefficient for experimentation. After researching optimization techniques, we discovered that caching the dataset could significantly reduce training time.

To implement this, we preprocessed all patient images and labels during the data loading phase and saved them as tensors to disk. This eliminated the need to reload and reprocess data for every training run. We wrote a utility function that either loads the cached tensors if they already exist, or creates and caches them if not.

Once cached, we loaded the tensors into ‘TensorDataset’ objects and created corresponding ‘DataLoader’ instances for the training, validation, and test sets. Each loader was assigned a batch size of 8, which allowed us to train the model efficiently while maintaining good memory usage. Below is a breakdown of the final dataset sizes:

- **Training samples:** 294 → **Training batches:** 37
- **Validation samples:** 37 → **Validation batches:** 5
- **Test samples:** 38 → **Test batches:** 5

6. Modeling Approach & Hyperparameter Tuning

1. Whole-Image Regression Pipeline

Our overall modeling pipeline followed a whole-image regression approach. Instead of performing pixel-wise segmentation, our model predicted the relative proportions of three tumor subtypes, Necrotic Core (p_1), Edema (p_2), and Enhancing Tumor (p_4), from individual 2D brain MRI slices.

2. Model Architecture: Modified VGG-16

We used a pre-trained VGG-16 convolutional neural network as the base architecture. Originally trained on over a million natural images from ImageNet, VGG-16 is composed of 13 convolutional layers grouped into 5 blocks (each followed by max-pooling) and 3 fully connected layers at the end for classification.

To adapt it for our regression task, we made the following modifications:

- **Input Layer:** Modified the first convolutional layer to accept 4-channel input, corresponding to the four MRI modalities (T1, T1CE, T2, FLAIR).
- **Output Layer:** Replaced the final classifier head with a custom regression head consisting of three dense layers.
- **Activation Function:** Used a sigmoid activation on the final layer to constrain outputs between 0 and 1, suitable for predicting proportions.

3. Training the Base Model & Manual Fine-Tuning

Since VGG-16 is a pre-trained model, we began by training only the regressor head, keeping the rest of the model frozen. This helped the model learn to adjust the new output layer to work well with the existing features before fine-tuning the rest of the network.

To identify an optimal hyperparameter configuration, we ran a grid search over 72 combinations involving learning rate, dropout rate, weight decay, optimizer choice, and number of epochs. The best configuration was logged as Experiment 0 in Table 1.

Following grid search, we conducted manual fine-tuning by unfreezing varying numbers of convolutional layers while also experimenting with values near the best base hyperparameters. This step aimed to adapt deeper feature representations to our medical imaging domain.

Below is the table that logs all of our experiments:

Base Model by Grid-Search + Fine-Tuning Summary							
Experiment	Unfrozen Layers	Learning Rate	Epochs	Dropout	Weight decay	RMSE	MAE
0	None	0.0001	15	0.5	0	0.2157	0.1738
1	4	0.0001	15	0.5	0	0.2248	0.1845
2	6	0.0001	15	0.5	0	0.2080	0.1626
3	6	0.0001	20	0.5	0	0.2132	0.1717

Base Model by Grid-Search + Fine-Tuning Summary							
4	6	0.0001	15	0.5	0.0001	0.2106	0.1680
5	8	0.0001	15	0.5	0.0001	0.2045	0.1574
6	10	0.0001	15	0.5	0.0005	0.2153	0.1698
7	10	0.0001	15	0.5	0.0001	0.2059	0.1594
8	10	0.0001	15	0.3	0.0001	0.2176	0.1680
9	10	0.0001	15	0.6	0.0001	0.2117	0.1675
10	10	0.0001	15	0.6	0	0.2089	0.1572
11	10	0.00001	10	0.6	0	0.2162	0.1648
12	8	0.00001	10	0.5	0.0001	0.2103	0.1592

Table 2. Summary of Grid Search and subsequent fine-tuning experiments

7. Model Evaluation & Results

To evaluate model performance, we used the following regression performance metrics:

- Mean Squared Error (MSE)
 - MSE measures the average squared difference between actual and predicted values. Because it squares the errors, it penalizes larger errors more heavily than smaller ones, making it sensitive to outliers.
 - $$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$
- Root Mean Squared Error (RMSE)
 - RMSE is commonly used in regression models to evaluate its accuracy and quantify its performance in predicting outcomes numerically. It calculates the average magnitude of the differences between the actual and predicted values.
 - $$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - y_i)^2}$$
- Mean Absolute Error (MAE)
 - MAE is commonly referred to as the scale-dependent accuracy because it considers observations from the same scale. It evaluates the average difference between the actual and calculated values.
 - $$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

These metrics were computed on both validation and test sets to assess prediction accuracy.

1. Training and Validation Loss Curves

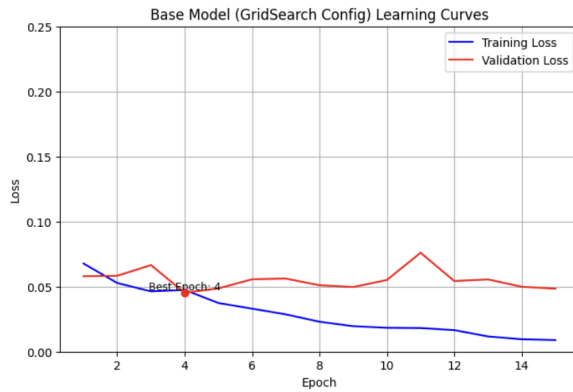


Figure 3. Loss Curves for Base Model

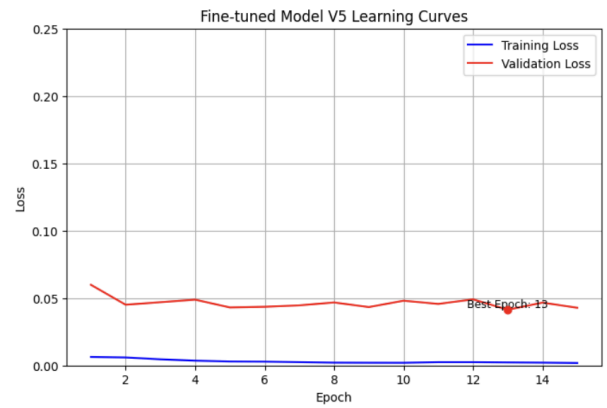


Figure 4. Loss Curves for Experiment 5

The best-performing base model, labeled Experiment 0 in Table 1, achieved an RMSE of 21.6% and an MAE of 17.4%. This configuration used a learning rate of 0.0001, a dropout rate of 0.5, and was trained for 15 epochs, with peak performance observed at epoch 4. As shown in Figure 3, both training and validation losses quickly stabilized early in training, suggesting that the model was able to adapt effectively to the task with minimal overfitting.

Following several rounds of manual fine-tuning, Experiment 5 emerged as the best-performing configuration. This version involved unfreezing 8 convolutional layers and a weight decay of 0.0001, allowing more of the pre-trained network to adapt to our medical imaging task. Every other hyperparameter remained the same as that of the base model.

It achieved an RMSE of 20.5% and an MAE of 15.7%, meaning the model's predictions were, on average, within ~16% of the actual tumor proportion values. This reflects strong predictive performance and suggests good generalizability across the dataset.

Although the loss curves may appear relatively flat, this is due to the narrow range of loss values, indicating consistently stable performance from the earliest stages of training, particularly after hyperparameters were optimized during the grid search.

2. Test Set Evaluation

Using the configuration from Experiment 5, we evaluated our model on the test set, which consisted of unseen data held out during training. The model achieved the following results:

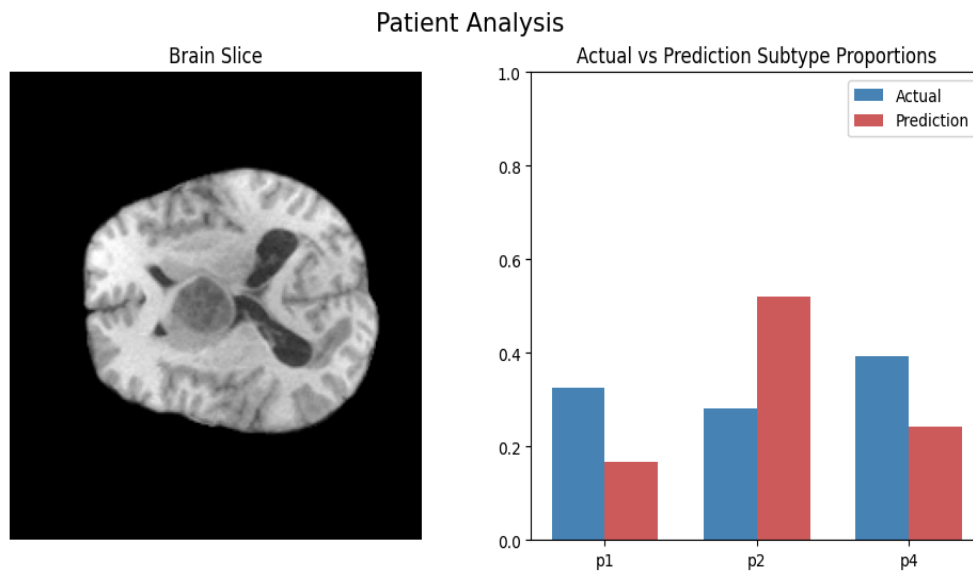
Evaluation Metrics for Testing Set		
MSE	RMSE	MAE
0.0876	0.2960	0.1976

Table 3. Final Evaluation Metrics of VGG-16 Model

While test error was slightly higher than on the validation set, the model still generalized moderately well, with predictions being on average 19.7% off from the true values. Although we initially aimed for an $MAE < 0.15$ and $RMSE < 0.20$, these preliminary results are promising and suggest that further tuning could bring us closer to our target.

3. Visual Demonstration

To illustrate what this model is learning, *Figure 5* shows predicted versus actual tumor subtype proportions for a randomly selected patient. The model successfully captured the general distribution, though it slightly overestimated Edema and underestimated both Necrotic Core and Enhancing Tumor proportions in this case.

**Figure 5.** Predicted vs. actual tumor subtype proportions for a randomly selected patient.

8. Code and Resource Organization

 Copy of Yet another copy of CSC509_Project_FinalVersion.ipynb

Our entire project pipeline was implemented within a single Jupyter notebook on Google Colab. This notebook contains all stages of the workflow, including:

1. Environment setup (see [Section 3: Dataset Overview](#))
2. Data preprocessing and loading (see [Section 5: Data Pre-Processing](#))
3. Model architecture and training (see [Section 6: Modeling Approach & Hyperparameter Tuning](#))
4. Evaluation and visualization (see [Section 7: Model Evaluation & Results](#))

To support collaboration and iterative experimentation, we adopted the following workflow:

- **Notebook Versioning through Duplicates**

Instead of working on a single shared notebook, which previously caused syncing and conflict issues, each team member created individual copies when running new experiments. This allowed us to experiment independently while preserving the integrity of the main pipeline. The notebook's title reflects this iterative, copy-based development process.

- **Modular Code Structure**

Key components of the pipeline (e.g., data normalization, caching, model definition, and plotting) were implemented as modular, reusable functions. This made the codebase easier to maintain and adapt during experimentation.

- **Dataset Caching for Efficiency**

To reduce load times and improve training efficiency, we cached preprocessed tensors to Google Drive. This significantly accelerated experimentation by avoiding repeated computation. Details on this process are provided in [Section 5.5: Data Caching](#).

- **Cloud-Based Resource Management**

Input data was accessed from a Google Cloud Storage (GCS) bucket using gcsfuse, while all cached files and metadata (e.g., the split log) were saved to Google Drive. This setup ensured persistent access and easy sharing across team members and Colab sessions.

9. Challenges & Problem Solving

One of the main challenges we encountered was the long training time for our model. Initially, without any form of data caching, training took approximately two hours, only for the process to fail due to an error. To address this, we implemented a data caching mechanism, which significantly reduced the training time and improved the efficiency of our workflow.

Another challenge involved limited storage capacity on our Google Drive, which prevented us from running extended training sessions for hyperparameter tuning. As a workaround, we opted for a more efficient search strategy by using Grid Search to identify suitable hyperparameters without exhaustive manual fine-tuning. Despite constraints, this method yielded reasonable results for our base model which allowed for a reduced scope for manual fine tuning later on.

We also faced limitations with our transfer learning setup using the VGG-16 model. Specifically, the model underperformed on the testing set, likely due to the small sample size. This highlights the importance of having sufficient and diverse data when fine-tuning deep learning models, especially for medical image regression tasks.

10. Future Direction & Learning Outcomes

The most immediate next step would be to apply data augmentation techniques to increase the diversity and effective size of our training dataset. Given that our current dataset is relatively small for training deep learning models, we aim to experiment with augmentation strategies to increase our sample size and evaluate their impact on model performance.

Another key direction is to explore transfer learning using models pre-trained on medical imaging datasets, such as RadImageNet or MedicalNet. While VGG-16 has shown utility in various domains, its original training on ImageNet (a natural image dataset) may limit its ability to extract features specific to medical images like MRI scans. Utilizing models trained on domain-relevant data could help improve feature extraction and overall predictive accuracy.

In addition, we plan to explore a broader range of hyperparameter tuning strategies. This includes integrating techniques like early stopping and Bayesian optimization, and potentially experimenting with different batch sizes, although the impact of the latter may be limited due to our dataset's size.

Overall, our results demonstrate the potential of this approach to support automated tumor characterization from MRI scans. While still in the early stages, models like ours could eventually assist clinicians by providing fast, quantitative assessments of tumor composition, serving as a complementary tool for more personalized treatment planning.

11. Individual Contributions & Reflection

References

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